

Generally, look for a proof by cases when there is no obvious way to begin a proof, but when extra information in each case helps move the proof forward.

Example 5: Formulate a conjecture about the final decimal digit of the square of an integer and prove your result. (Beautiful)

Solution: The smallest perfect squares are 1, 4, 9, 16, 25, 36, 49, 64, 81, 100, 121, 144, 169, 196, 225, and so on. We notice that the digits that occur as the final digit of a square are 0, 1, 4, 5, 6 and 9 with 2, 3, 7 and 8 never appearing as the final digit of a square.

We conjecture this theorem: The final decimal digit of a perfect square is 0, 1, 4, 5, 6 or 9. How can we prove this theorem?

We first note that we can express an integer n as $10a+b$, where a and b are positive integers and b is 0, 1, 2, 3, 4, 5, 6, 7, 8 or 9. Here a is the integer obtained by subtracting the final decimal digit of n from n and dividing by 10.

Next, note that $(10a+b)^2 = 100a^2 + 20ab + b^2 = 10(10a^2 + 2b^2) + b^2$, so that the final decimal digit of n^2 is the same as that of b^2 .

Furthermore, note that the final decimal digit of b^2 is the same as the final decimal digit of $(10-b)^2 = 100 - 20b + b^2$. Consequently, we can reduce our proof to the consideration of 6 cases:

Case 1: The final digit of n is 1 or 9. Then the final decimal digit of n^2 is the final decimal digit of $1^2 = 1$ or $9^2 = 81$, namely 1.

Case 2: The final digit of n is 2 or 8. Then the final decimal digit of n^2 is the final decimal digit of $2^2 = 4$ or $8^2 = 64$, namely 4.

Case 3: The final digit of n is 3 or 7. Then the final decimal digit of n^2 is the final decimal digit of $3^2 = 9$ or $7^2 = 49$, namely 9.